

#Practical – 1 Facebook

1. Import library and dataset and print DS

- import pandas as pd
- df = pd.read_csv("facebook.csv",sep=";")
- df.head()

2. Create Subset

- df1 = df[["Type","Category"]]
- df1.head()
- df2 = df[["Type","like","Paid"]]
- df2.head()

3. Merge Dataset

- merge = pd.merge(df1,df2,on="Type")
- merge.head()

4. Sort Data

- sorted = df.sort_values(by="like",ascending=False)
- sorted.head()

5. Shape & Reshape Data

- df.shape
- transpose = df.T
- transpose.head()
- import numpy as np
- arr = np.array([5,3,6,9,8,7])
- new_arr = arr.reshape(2,3)
- new_arr

6. Other IMP functions :-

- df.info() → Displays **summary of the dataset structure** including:
- df.describe() → Provides **statistical summary of numerical columns**, including:
- df.isnull().sum() → Shows the **total number of missing (null) values in each column**.
- df.fillna(0, inplace=True) → Replaces all **missing (NaN) values with 0** in the dataset.

Practical 1 Viva Questions (Facebook Dataset)

Basic Questions

1. What is pandas?

 Pandas is a Python library used for **data manipulation and analysis**. It provides data structures like DataFrame and Series.

2. What is a DataFrame?

 A DataFrame is a **2-dimensional labeled data structure** with rows and columns (like a table in Excel).

3. What is CSV file?

 CSV (Comma Separated Values) is a file format used to store **tabular data in plain text form**.

Dataset Loading

4. Why do we use read_csv()?

 It is used to **load dataset from a CSV file into a DataFrame**.

5. What does sep=";" mean?

 It means the data is separated using **semicolon (;) instead of comma (,)**.

Data Exploration

6. What does df.info() do?

 It gives **summary of dataset structure, data types, and missing values**.

7. What is the use of df.describe()?

👉 It gives **statistical summary of numerical columns** like mean, min, max, etc.

8. What is `df.isnull().sum()` used for?

👉 It is used to **find missing values in each column**.

9. Why do we handle missing values?

👉 Because missing values can **affect accuracy of analysis or machine learning models**.

📌 Data Manipulation

10. What is subset in pandas?

👉 A subset is a **part of dataset containing selected columns or rows**.

11. Why do we create subsets?

👉 To **focus on required data only and reduce complexity**.

12. What is merging in pandas?

👉 Merging combines **two DataFrames using a common column (key)**.

13. What is the use of `merge()` function?

👉 It is used to **combine datasets based on a common column**.

14. What is sorting in pandas?

👉 Sorting arranges data in **ascending or descending order**.

15. What does `sort_values()` do?

👉 It sorts DataFrame based on a **specific column**.

📌 Data Structure Questions

16. What is transpose (df.T)?

👉 It **converts rows into columns and columns into rows**.

17. What is shape of DataFrame?

👉 df.shape returns **number of rows and columns**.

18. What is numpy used for?

👉 NumPy is used for **numerical operations and array manipulation**.

19. What is reshaping in numpy?

👉 Reshaping changes the **dimension of an array without changing data**.

20. Why do we use reshape?

👉 To convert data into required **matrix form for processing**.

⚡ Advanced Viva (sometimes asked)

21. Difference between DataFrame and Series?

- Series → 1D data
 - DataFrame → 2D table
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22. Difference between merge and concat?

- merge → based on key column
- concat → joins datasets vertically or horizontally

23. Why is pandas important in data science?

👉 Because it helps in **data cleaning, analysis, and manipulation easily**